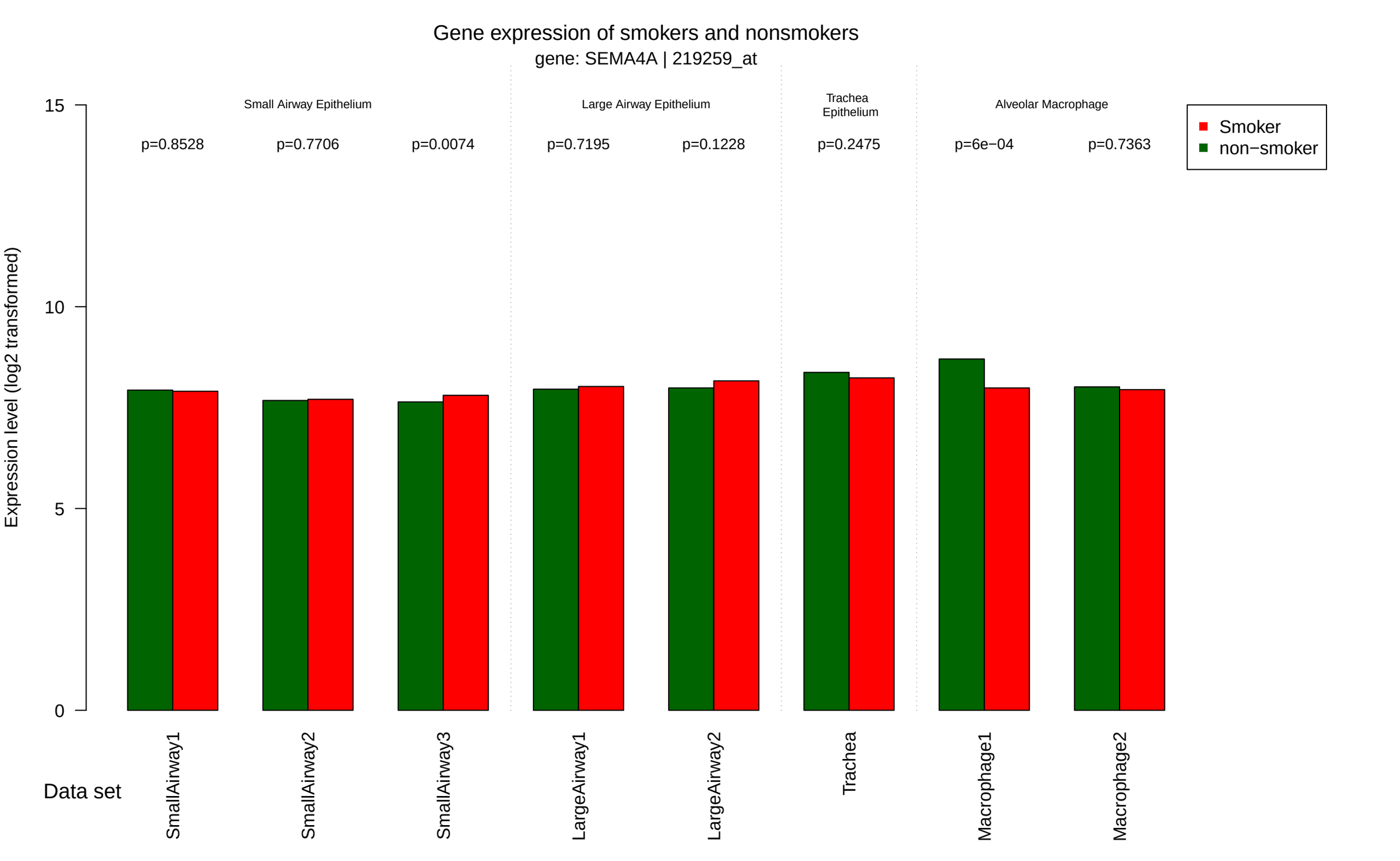
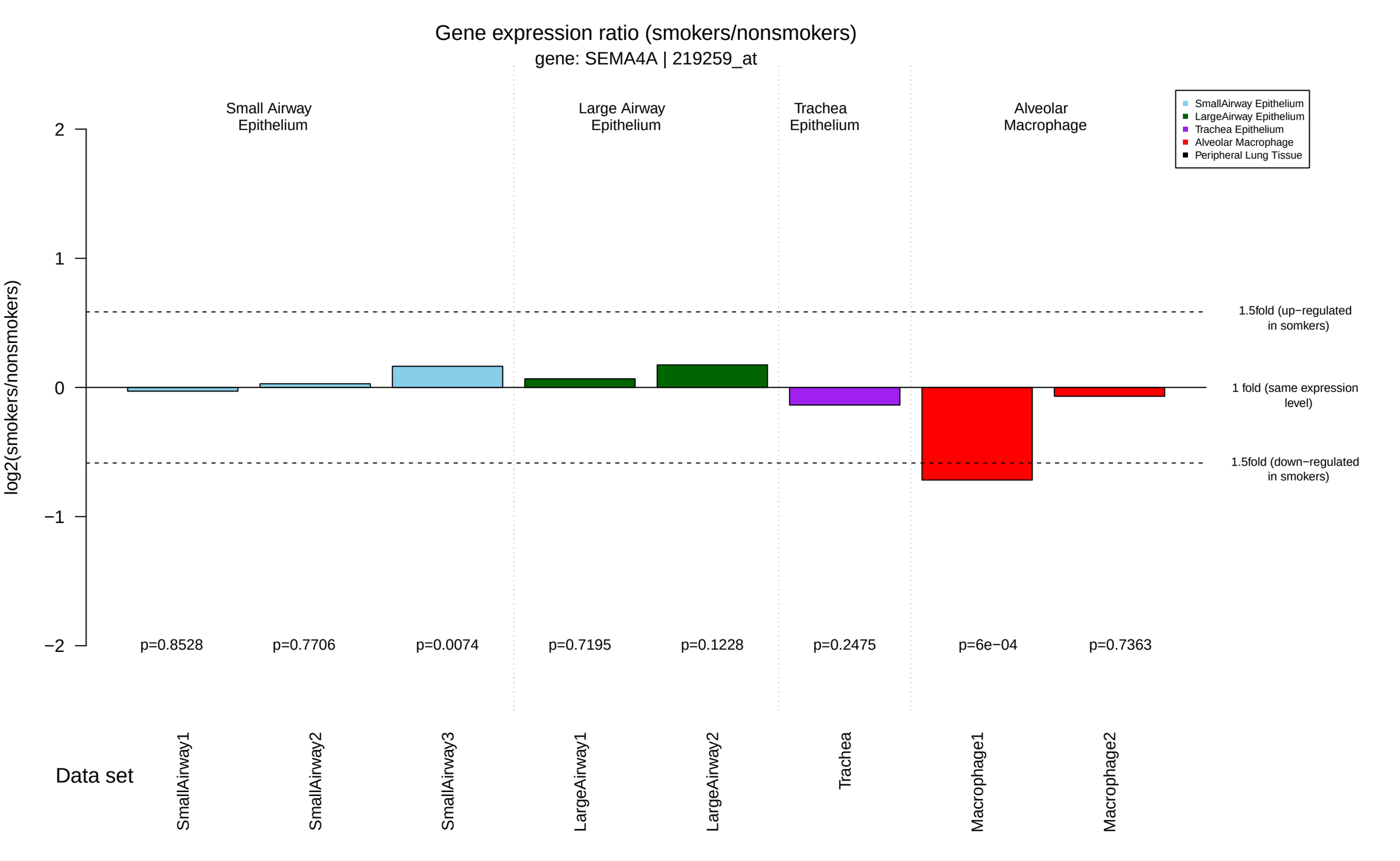
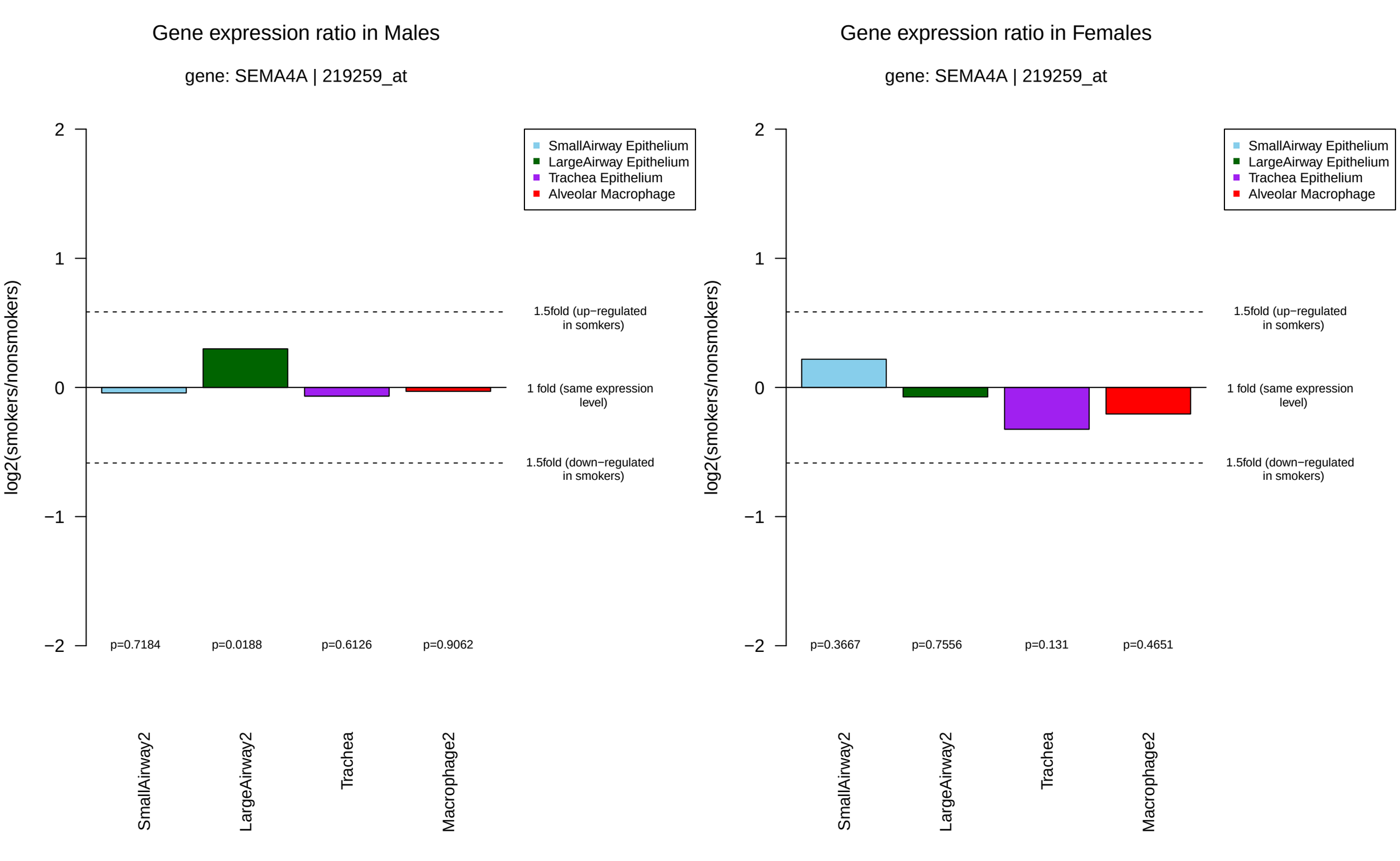
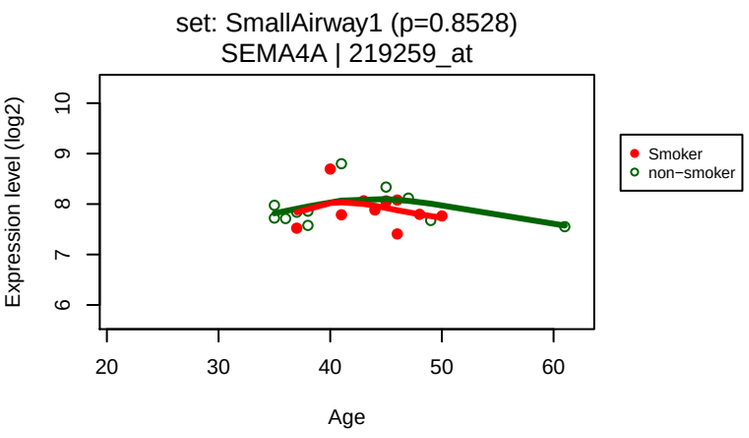
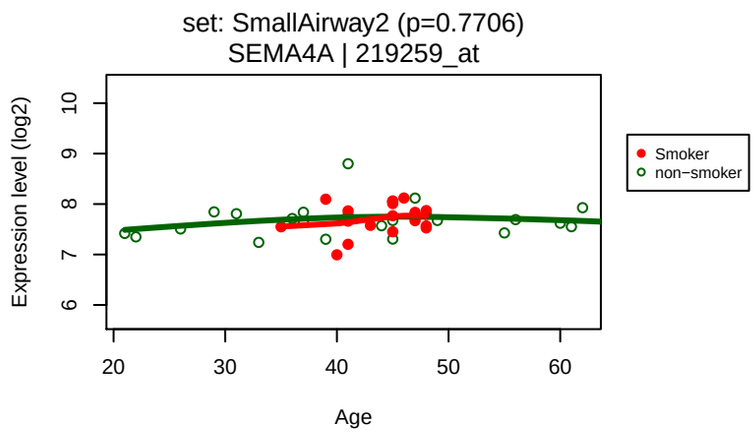
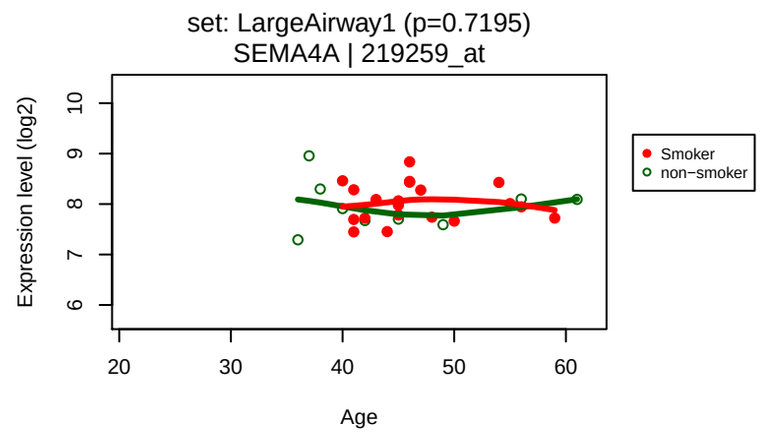

Gene expression of smokers and nonsmokers
gene: SEMA4A | 219259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8528
p=0.7706
p=0.0074
p=0.7195
p=0.1228
p=0.2475
p=6e−04
p=0.7363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

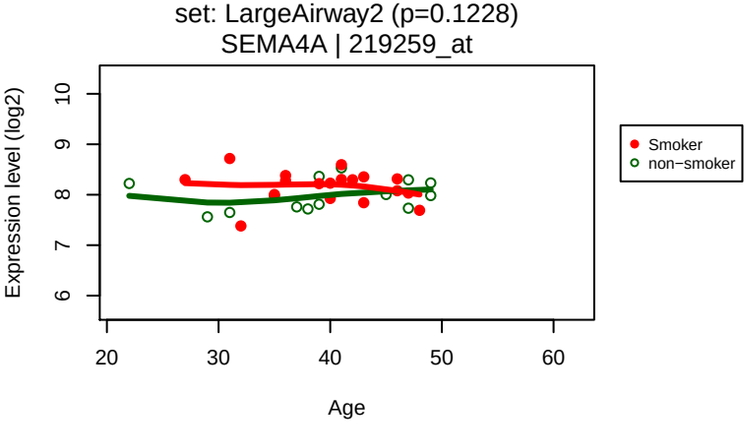

Gene expression ratio (smokers/nonsmokers)
gene: SEMA4A | 219259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8528
p=0.7706
p=0.0074
p=0.7195
p=0.1228
p=0.2475
p=6e−04
p=0.7363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

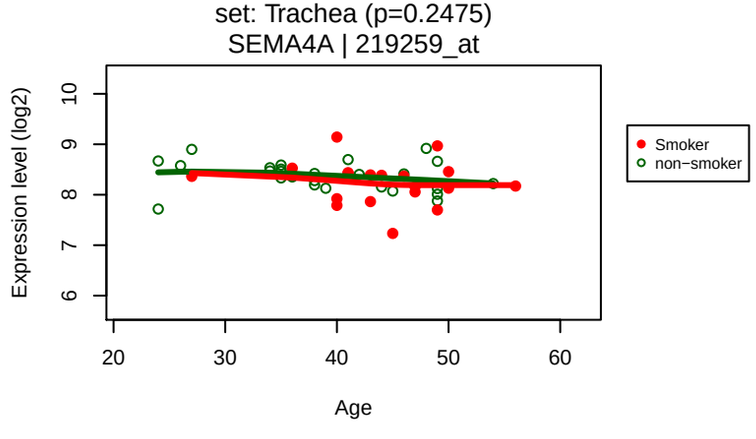

Gene expression ratio in Males
gene: SEMA4A | 219259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7184
p=0.0188
p=0.6126
p=0.9062
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA4A | 219259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3667
p=0.7556
p=0.131
p=0.4651
SmallAirway2
LargeAirway2
Trachea
Macrophage2

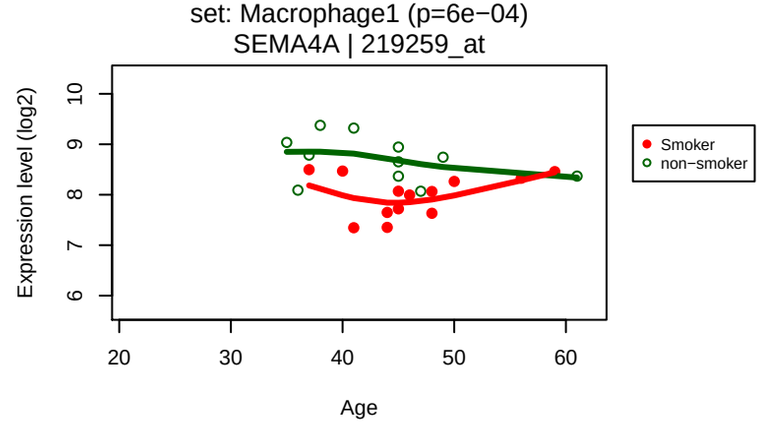

set: SmallAirway1 (p=0.8528)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker

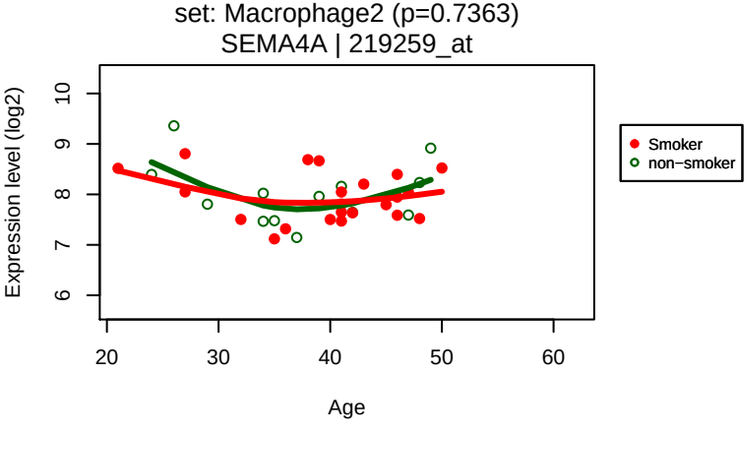

set: SmallAirway2 (p=0.7706)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker

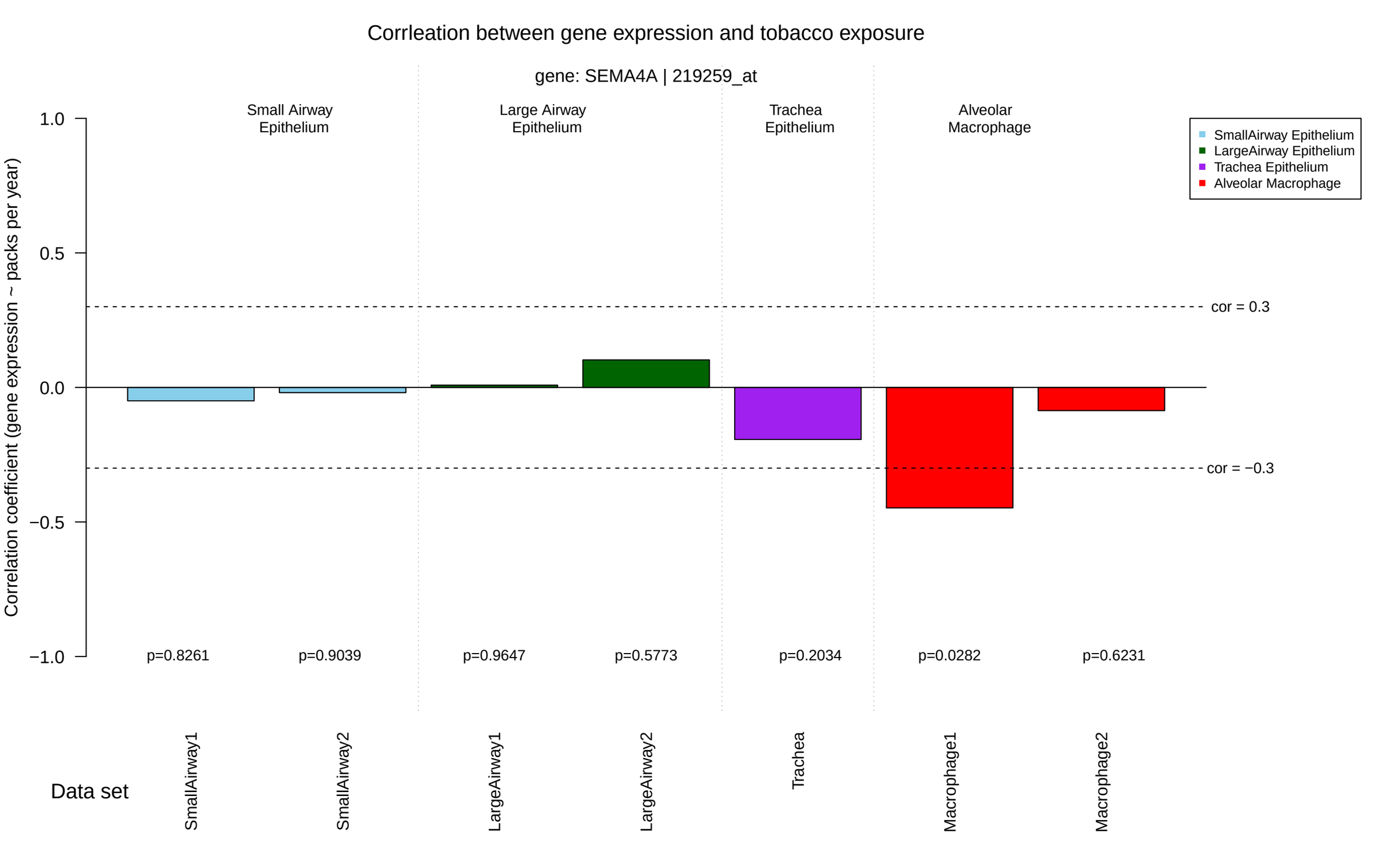

set: LargeAirway1 (p=0.7195)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1228)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2475)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=6e−04)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7363)
SEMA4A | 219259_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA4A | 219259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8261
p=0.9039
p=0.9647
p=0.5773
p=0.2034
p=0.0282
p=0.6231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2